

Resonance Bastion — animation and interaction refinement

2026-10-07 · build 0.2.2-animation-refinement · simulation rb-sim-v2. This is the current development checkpoint. The earlier full review remains the 0.2.0 snapshot. The blueprint and full intended game scope are preserved.

What changed

Damage marks now follow the current body pose, including stride/action offsets, and remain within its alpha silhouette. A shared precomputed atlas avoids per-unit, per-frame compositing. All 1,792 actual body/band pairs passed pixel, anchor, gutter and packed-image checks. Original source images were not edited; the raw idle-overlay audit still reports its historical 427 mismatches.

Walking cadence follows actual distance travelled, slowing or freezing when motion is slowed or blocked. Dash does not advance walking phase. Ability cast-to-release frames continue instead of restarting; dash poses clamp and Repair Node channel animation no longer resets at every healing pulse. These fixes improve continuity; they do not certify anatomically correct foot plants.

Ranged units previously could stop within weapon range behind an obstruction, clip a cover corner with a finite-size projectile, or repeatedly cancel warmup because an expired stuck watchdog never re-armed. New v2 firing routes require clearance for the blueprint's 0.05 GU kinetic radius, finish alignment inside the chosen firing cell and re-arm the watchdog. Independent blocked-Core and corner fixtures verify engagement. Perception still uses its own sight rules.

The guided practice Mine moved from (22,30) to (18,24), based on measured enemy trajectories and a controlled placement comparison. It now triggers and causes actual damage in the canonical tutorial; the 18-tick warning remains intact. This changes only new disposable guided clones, not the player's fortress or a stored checkpoint. No weapon damage, armor, HP, speed, price or economy values were tuned.

Compatibility and verification

New practice uses rb-sim-v2. Retained v1 checkpoints keep their stored plan and army, and replay the original exact 6,150-tick result/hash. Saved results and campaign versions are not silently rewritten. Practice still awards no real currency/XP and causes no persistent campaign injury.

Strict types, lint, production build and 46 unit tests passed. The full 19-test production-browser suite passed after the renderer/AI changes. The tutorial browser cases were rerun after the guided Mine placement. Checks include native checkpoint deletion/reload, saved retry paths, context loss, pause/speed, reflow/input and equal enabled/muted combat outcomes. The canonical simulation matches across tick groupings 1/2/4/8/12 with 24 kills and 30 TP.

Current canonical result: Victory, 6,879 ticks (114.65 combat seconds), Core 10,000; snapshot SHA-256: 1fc32c1de93483123a91b0b018aa0f90a36e508c406acc27b14f2990016ba8b6.

Legacy v1 SHA-256: 8961c7817d3d73989a00c3e377387e6dde1ebbee16d219d5c397afc2be7839c7. These hashes describe the fixed UUID fixture, not every player's owned-asset IDs.

The overlay atlas is 2,048×274 with 414 unique rectangles. All loaded texture pixels total 5,549,056 within the existing 6,000,000 budget. The recorded atlas build took about 0.79 seconds on this desktop environment; it is not an Android load-time or frame-rate measurement.

Measured tutorial comparisons

Twelve deterministic cases were measured with a fixed schedule and authored UUIDs: baseline, former Mine position, role omissions, half-health conditions, Bulwark alone and an unarmed diagnostic. Eleven reach Victory; the unarmed fixture reaches the intended Inevitable Defeat rule while its Core is still alive. These are mechanism/sensitivity comparisons, not whole-game balance approval.

The healthy baseline needs no repair. Removing Sentries exercises 593.76 actual healing; half-health Sentries exercise 600. The new Mine causes 100 actual body loss in baseline; its former position causes zero. Actual loss is capped by the recipient's remaining HP, rather than reporting theoretical blast damage.

Moving the Mine does not guarantee a shorter battle: the former placement finishes in 109.52 seconds versus 114.65 now. Different kills and obstructions change later targeting/action/random-consumption order. Role removal can also renumber runtime IDs and alter tie-breaks. Do not infer an isolated DPS ranking or nerf a support unit from these single-layout outcomes. Before-v1 versus v2 also changes the version-seeded accuracy stream; independent geometry tests, rather than those aggregate totals alone, establish the route fixes.

Machine evidence: docs/evidence/POSE_DAMAGE_RUNTIME.json, TUTORIAL_BALANCE.json and TUTORIAL_BALANCE_BEFORE.json. Research and primary references: docs/RESEARCH_ANIMATION.md.

What still needs work

The existing broad source-rig slices remain candidate art. Explicit per-asset joints, eight-direction ground contacts, weapon/tool sockets, moving-and-firing blends, accurate falls and consistent equipment still need authored refinement. Silhouette clipping can prevent detached damage pixels; it cannot prove that underlying limbs, gear or downed poses are correct. The per-asset acceptance matrix in ANIMATION_AUDIT.md remains applicable.

The new four-key Rifle study is a source guide only, generated from the retained Rifle reference. It has a magenta backdrop and is not loaded into the game. The first repeated-pose attempt was rejected. The new guide still needs consistent native alignment, eight distinct phases/directions and contact/equipment checks before it can become a runtime animation. No mass roster production or final visual acceptance is claimed.

The video records actual production UI at 1x, 800×1280, for roughly 45 seconds of combat, including the Mine interaction. It contains no captured system audio and is not a full-battle proof. Desktop Chromium cannot establish Lenovo tablet frame pacing, heat/memory behavior, TalkBack or touch responsiveness.

Next development and useful input

Continue per-asset articulation/contact authoring and focused live scrutiny, then the Gate3 persistent campaign loop: atomic results/rewards/injuries, repair/restoration/progression and backup recovery. Preserve practice isolation and all later blueprint content. No new gameplay decision is blocking engineering.

Useful user feedback is visual: whether the live motion and Rifle keypose design match the intended look, and any visible sliding, gear drift or unreadable effect on the tablet. Exact Android/browser version and RAM help reproduce later device measurements. A static PDF/video review alone cannot certify physical performance.

[Open/download the actual 1x motion video](#)

Controlled tutorial comparisons

Case	Outcome	Seconds	Core HP	Repair	Mine loss
authored-baseline	Victory	114.65	10000.00	0.00	100
former-mine-position	Victory	109.52	10000.00	0.00	0
no-repair	Victory	105.58	10000.00	0.00	100
no-mine	Victory	109.52	10000.00	0.00	0
no-barrier	Victory	99.45	10000.00	0.00	73
one-sentry	Victory	119.97	10000.00	0.00	100
no-sentries	Victory	121.35	9889.58	593.76	100
no-rifle	Victory	110.05	9591.66	150.00	100
rifle-half-health	Victory	137.93	9983.33	0.00	100
sentries-half-health	Victory	114.65	10000.00	600.00	100
warden-alone	Victory	101.52	8629.15	0.00	0
unarmed-core	Defeat	12.98	10000.00	0.00	0

Actual capped damage/healing, not theoretical DPS. Fixed Bulwark/Bastion schedule and layout; role removal changes later targeting, runtime tie-breaks and random consumption. Do not infer whole-game balance or isolated role rankings. The unarmed diagnostic ends under Inevitable Defeat, not Core destruction.

Pose attachment inspection

Left: original idle damage marks. Right: current pose attachment. Native pixels enlarged by the read-only browser checker; bodies and source PNGs unchanged. This validates overlay geometry, not final anatomy.

Resonance Bastion — pose attachment comparison

Native atlas pixels, 4× nearest-neighbor. Left: earlier idle marks. Right: corrected pose attachment. Diagnostic sheets; no natural-motion approval inferred.

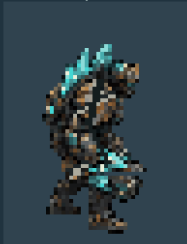
friendly.rifle_squad — move frame 2, face 2

Earlier idle marks	Corrected pose marks
	

enemy.runner — move frame 2, face 2

Earlier idle marks	Corrected pose marks
	

enemy.raider — move frame 2, face 2

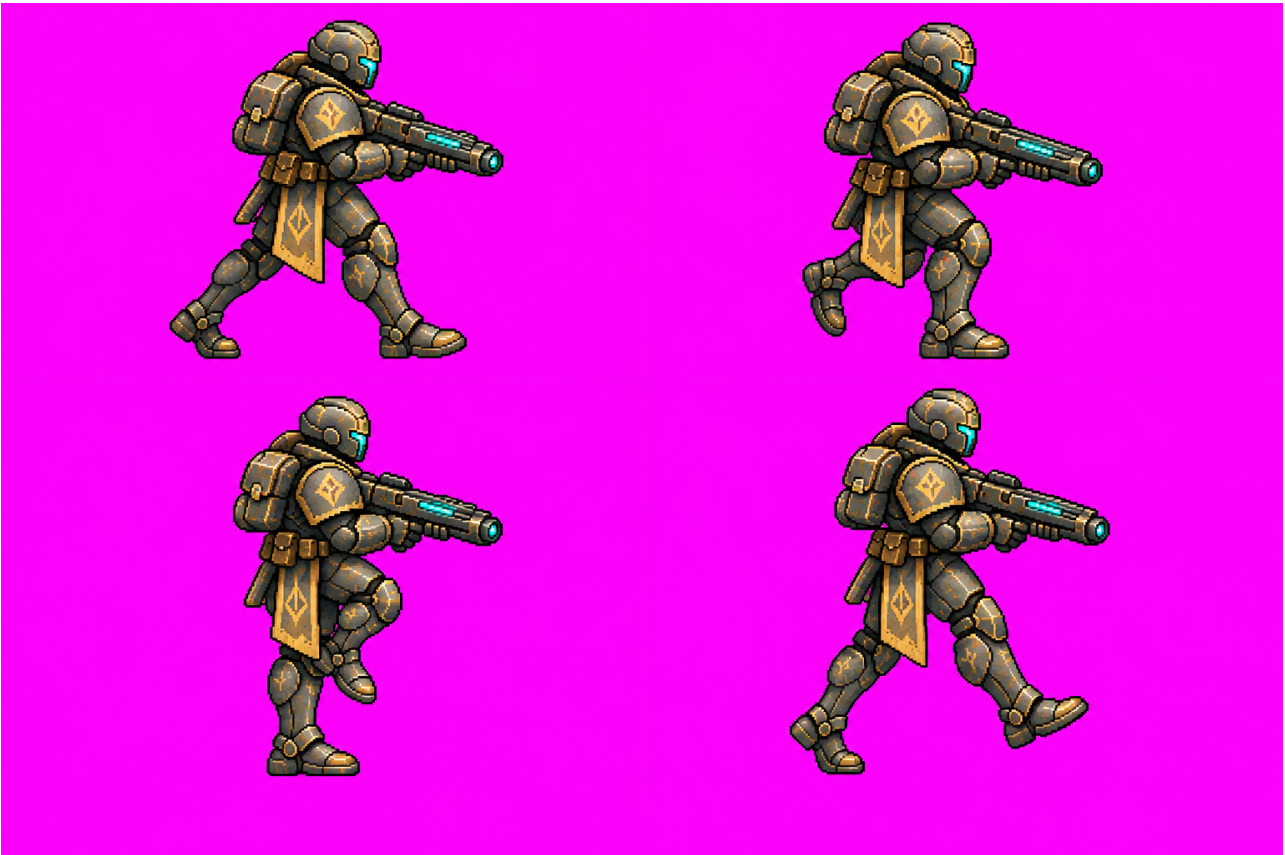
Earlier idle marks	Corrected pose marks
	

warden.bulwark — move frame 2, face 2

Earlier idle marks	Corrected pose marks
	

Rifle keypose source guide — not in game

Four conceptual keys: contact/down/passing/up. Magenta backdrop is source-study presentation. Not transparent native sprites, a finished eight-phase cycle, eight directions or approval of anatomy/equipment. Review this design before detailed runtime authoring.



Useful feedback: does this design and weight match the intended Rifle, and what looks wrong in the live motion? Exact Android/browser version and RAM will help physical-device measurement. Engineering is not blocked by a new gameplay-rule question.